

REAL ANALYSIS GENERAL EXAM AUGUST 18, 2025

- Solve the following five questions. The exam is 2 hours long.
- To receive full credit, you must clearly state which theorems you are using in your proofs, and your proofs must be fully justified and explained.

Question 1

Let m be the Lebesgue measure on $\mathbb{R}$ and let E be a Lebesgue measurable set such that $m(E) < \infty$. Let $f_n, f : E \rightarrow \mathbb{R}$ be measurable functions such that $f_n \rightarrow f$ almost everywhere. Assume that

$$\int_E |f_n|^3 dm \leq 1, \text{ for any } n \in \mathbb{N}.$$

Prove that

$$\lim_{n \rightarrow \infty} \int_E f_n dm = \int_E f dm$$

Question 2

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f : X \rightarrow [0, \infty]$ be a non-negative $\mathcal{M}$ -measurable function on X . Let

$$T = \{t \in [0, \infty] : \mu(f^{-1}(\{t\})) > 0\}.$$

Prove that if f is integrable, then T is a countable set.

Question 3

Let $\mathcal{M}$ denote the Lebesgue measurable sets on $\mathbb{R}$ and let m be the Lebesgue measure on $\mathcal{M}$ and τ be the counting measure on $\mathcal{M}$. Show that $m \ll \tau$ but $\frac{dm}{d\tau}$ doesn't exist. Explain why this doesn't contradict with the Radon-Nikodym Theorem?

Question 4

Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space with $\mu(X) = \infty$.

(a) Show that there exists a disjoint sequence $\{E_n\}_{n \in \mathbb{N}} \subset \mathcal{M}$ (i.e., $E_n \cap E_m = \emptyset$, when $n \neq m$), such that $\bigcup_{n \in \mathbb{N}} E_n = X$ and $\mu(E_n) \in [1, \infty)$ for every $n \in \mathbb{N}$.

(b) Show that there exists an $\mathcal{M}$ -measurable function $f : X \rightarrow [0, \infty]$ such that $f \in L^p(X, \mathcal{M}, \mu)$ for all $p \in (1, \infty]$ and $f \notin L^1(X, \mathcal{M}, \mu)$.

Question 5

(a) Let $f \in L^1(\mathbb{R}^n, m)$. Prove that the set of points where the maximal function is finite, $\{x \in \mathbb{R}^n : Mf(x) < \infty\}$, has full measure (i.e., its complement is a set of measure zero). The Hardy-Littlewood maximal function is defined as:

$$Mf(x) = \sup_{r>0} \frac{1}{m(B(x, r))} \int_{B(x, r)} f(y) dm(y).$$

(b) Give the definition of the Lebesgue set and state the Lebesgue Differentiation Theorem. Give an example of a function f that is in $L^1_{\text{loc}}(\mathbb{R}, m)$ but whose Lebesgue set is not all of $\mathbb{R}$. Justify your answer.